



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology
Department of Computer Science
CSC 3215: Web Technologies

Semester: Summer 2025-26

Group No:7

Section:L

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Project Title

Medicare-Hospital Management System

Short Description

Medicare is a web-based hospital management system that streamlines appointments, patient queues, medical follow-ups, and department management. It focuses on reducing manual paperwork, preventing scheduling conflicts, and providing real-time operational transparency for hospital operations.

Introduction

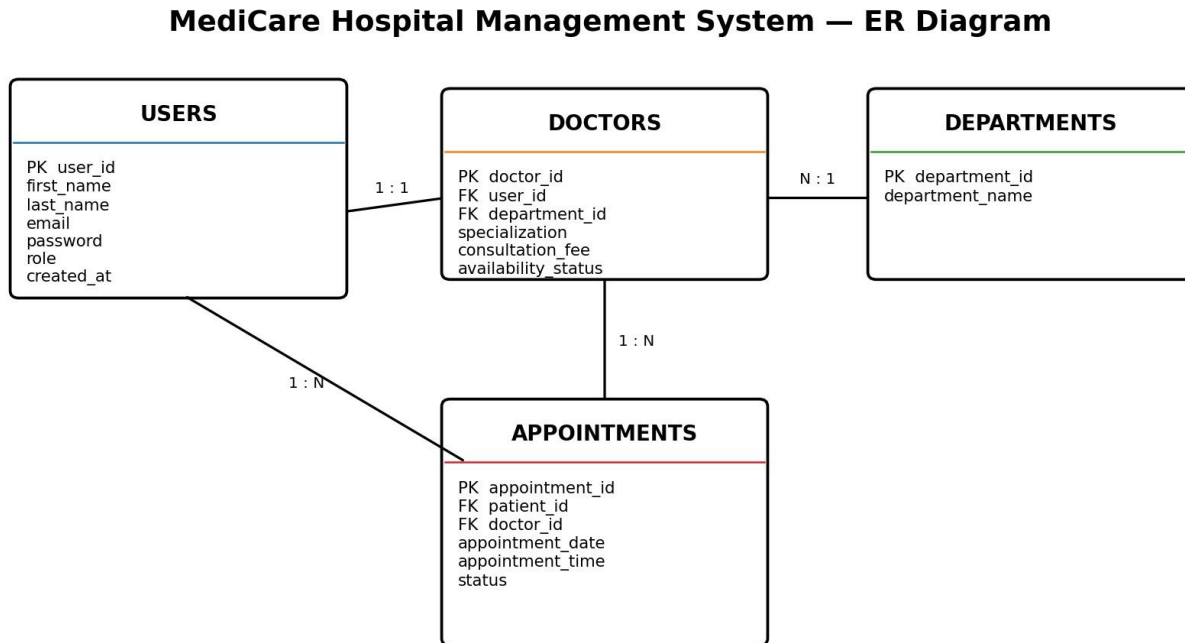
Medicare digitizes hospital operations to reduce waiting times and scheduling conflicts using PHP, MySQL, JavaScript. The platform enables patients to book appointments seamlessly while empowering doctors and staff to handle daily queues and medical records in real time.

Background Study

Traditional paper-based hospital management causes long patient waiting times, appointment overlaps, and lost medical records. Manual record-keeping makes it difficult for administrators to monitor hospital performance and resource allocation efficiently. But Modern web-based systems solve these issues by centralizing operations. By implementing role-based access for Admins, Doctors, Receptionists, and Patients, the system automates online appointment booking, live queue monitoring, and digital record tracking—eliminating administrative delays and improving overall healthcare delivery.

ER Diagram

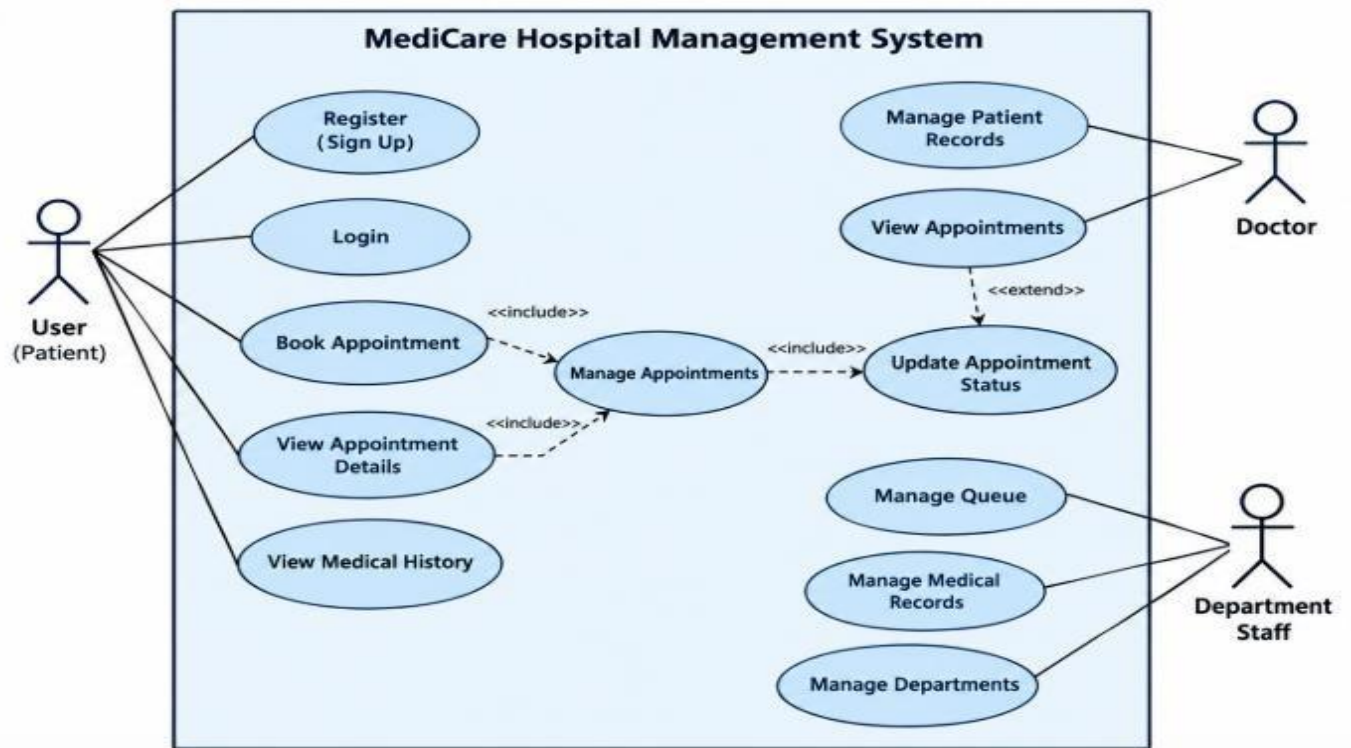
The following ER diagram represents the core database entities used in the MediCare system.



Core entities reflect the implemented project structure: users, doctors, departments and appointments.

Use Case Diagram

The following use case diagram shows the main interactions of Admin, Doctor, Receptionist, and Patient with the system.



UI (Figma/Sketch)

The following wireframe presents the tentative dashboard/UI structure based on the implemented MediCare system.

UI (Figma/Sketch)

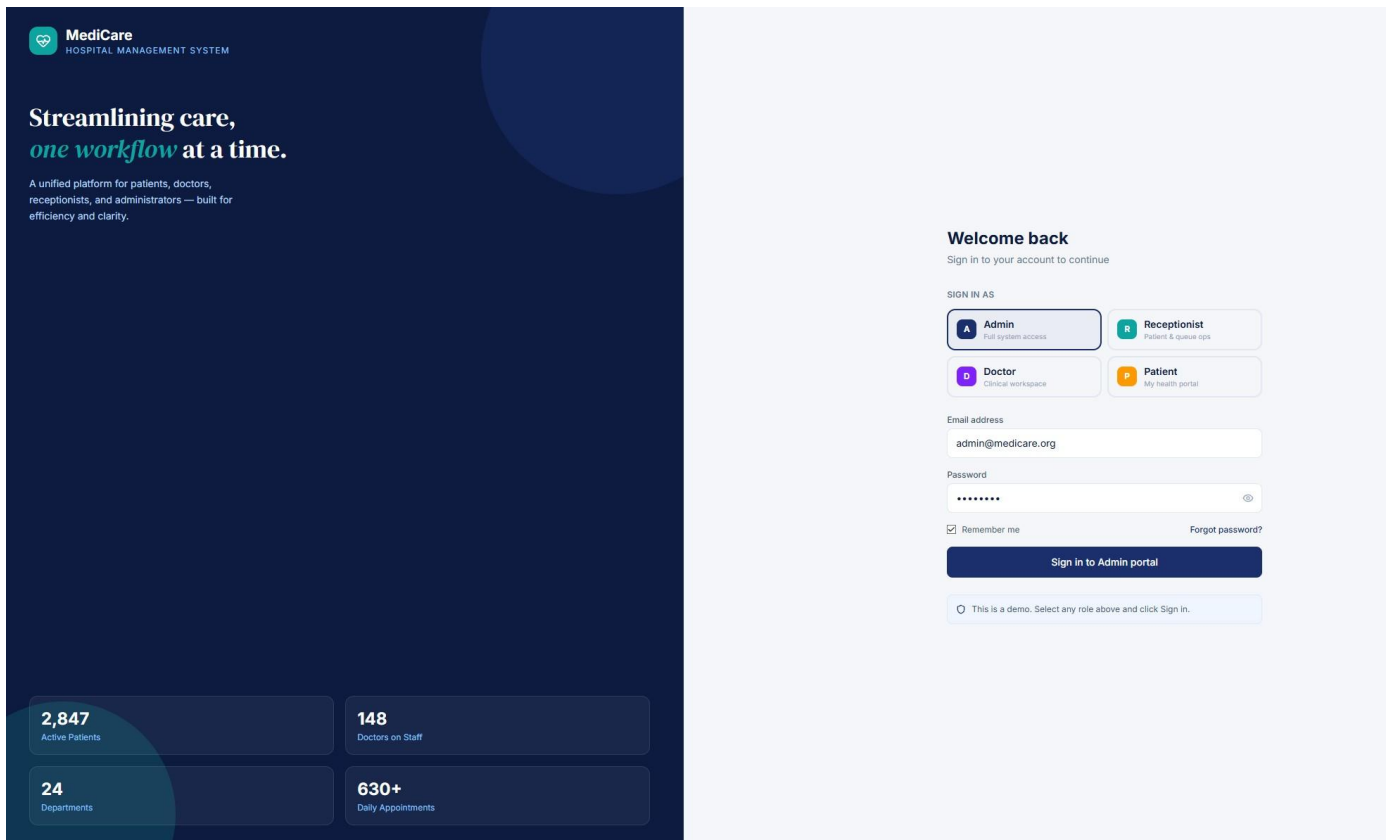


Fig1: Login and Signup page

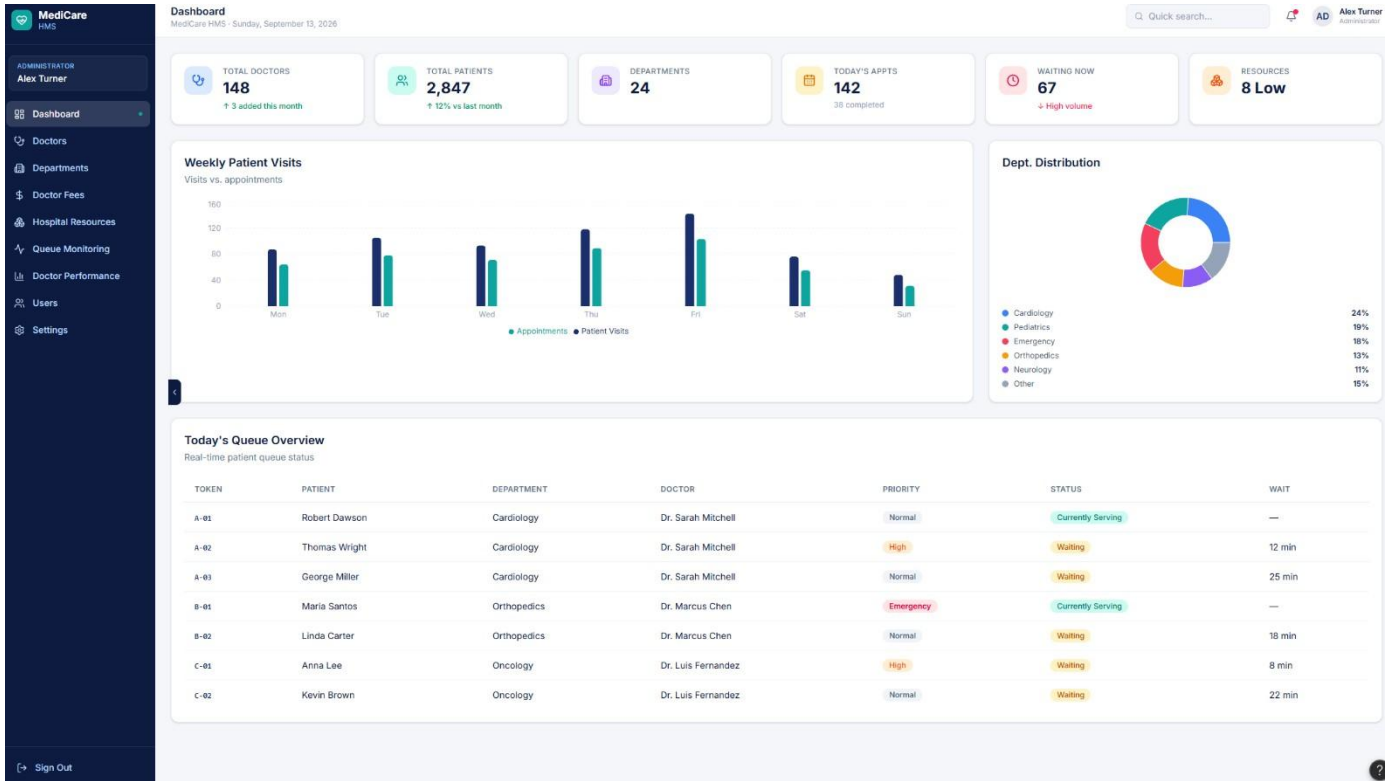


Fig: Dashboard Of Users

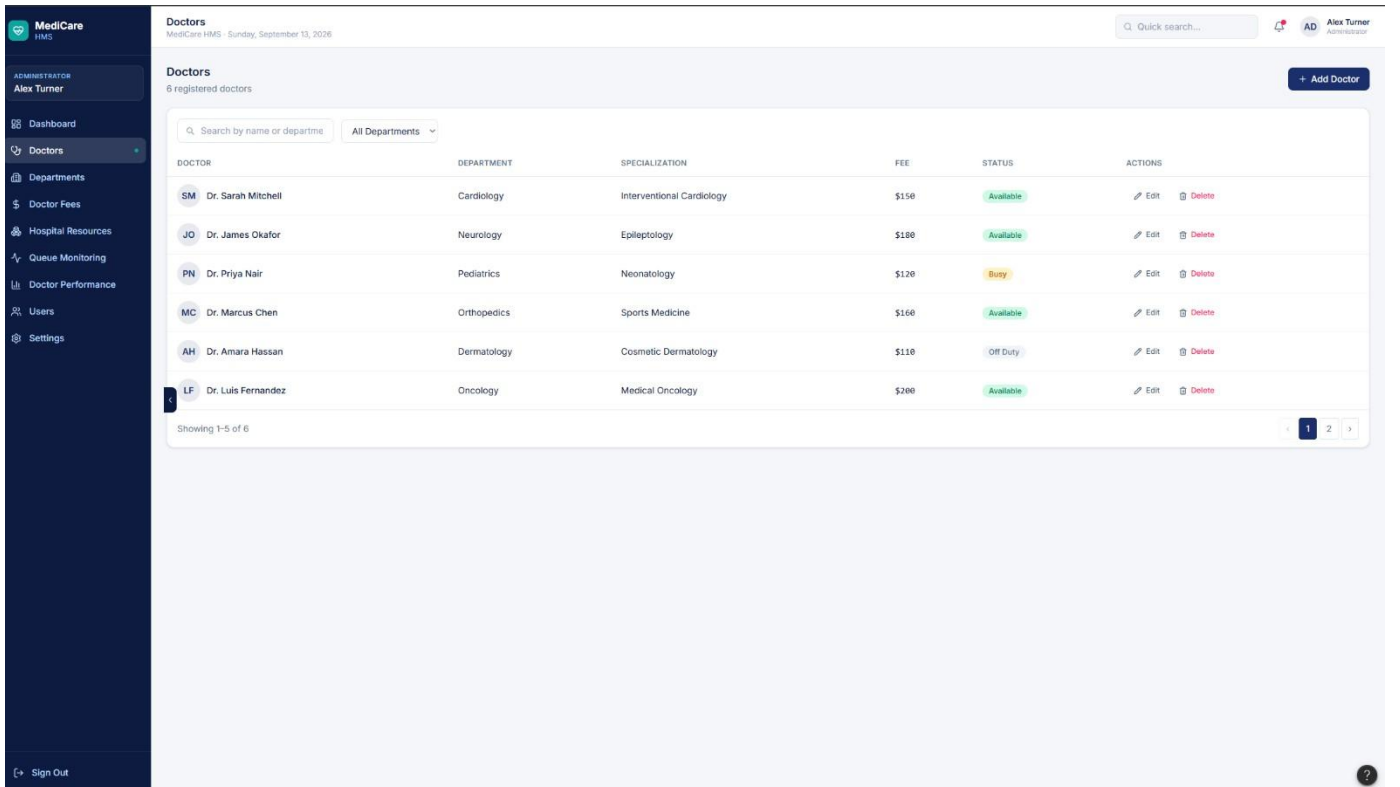


Fig: Doctors Section

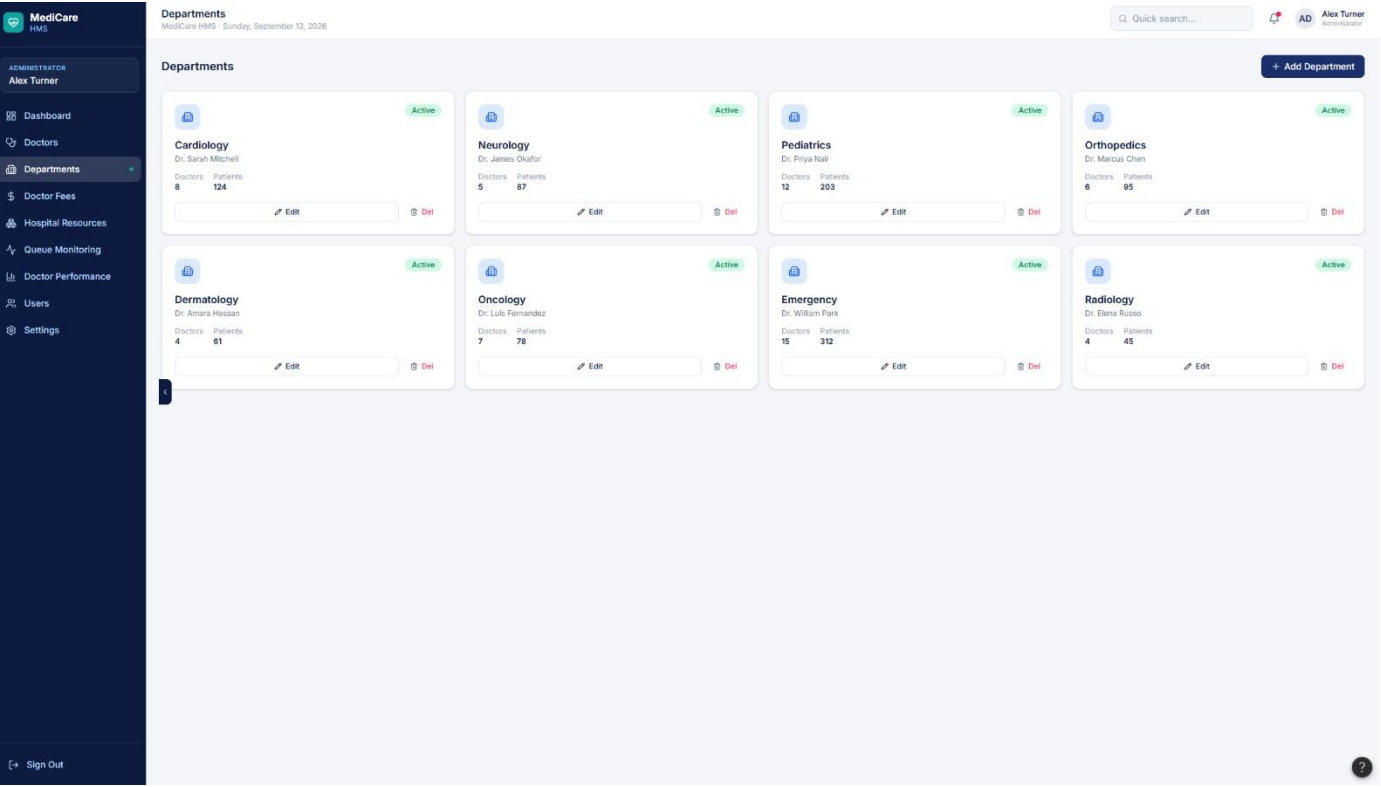


Fig: Department Section

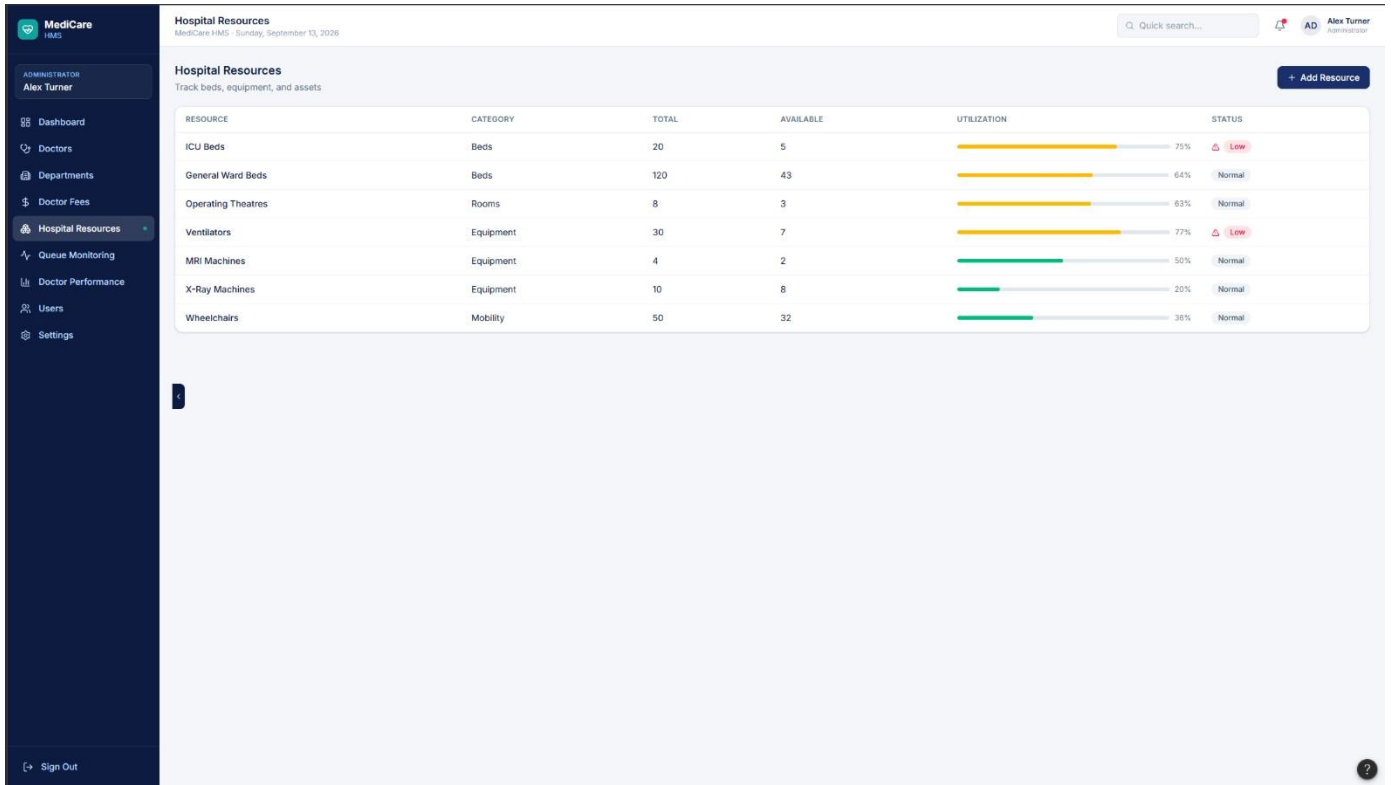


Fig: Hospital Resources View

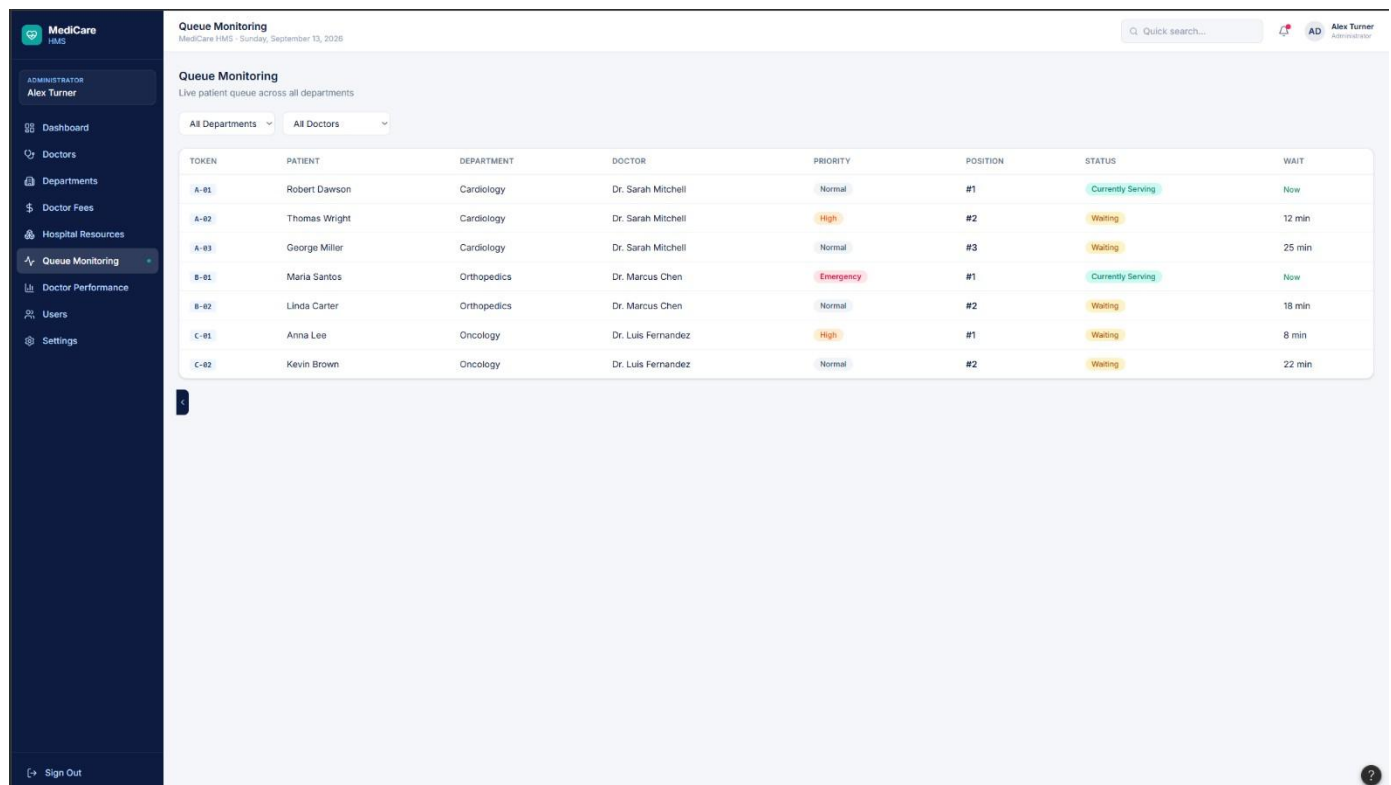


Fig: Queue Monitoring View

MediCare
HMS

ADMINISTRATOR
Alex Turner

Dashboard

Doctors

Departments

Doctor Fees

Hospital Resources

Queue Monitoring

Doctor Performance

Users

Settings

Sign Out

Users

MediCare HMS - Sunday, September 13, 2026

Quick search...

AD Alex Turner
Administrator

Users

All system users

+ Add User

NAME	EMAIL	ROLE	STATUS	JOINED	ACTIONS
Alex Turner	admin@medicare.org	Admin	Active	Jan 2024	Edit
Sarah Johnson	receptionist@medicare.org	Receptionist	Active	Mar 2024	Edit
Dr. Sarah Mitchell	smitchell@medicare.org	Doctor	Active	Feb 2023	Edit
James Wilson	patient@medicare.org	Patient	Active	Jun 2025	Edit

Fig: Users View

Project implementation alignment: The proposal uses the same four role structure and core modules developed for MediCare, including authentication, dashboards, departments, doctors, appointments, queue-related functions, and role-specific operations.

Types of Users: 1. Admin 2. Doctor 3. Receptionist 4. Patient

Common Features (Available to All Users)

- **Authentication**
 1. Login to the system
 2. Logout from the system
 3. User registration
- **Account Management**
 1. Change or reset password
 2. Manage profile information (view, edit, delete)
- **Dashboard**
 1. Access a personalized dashboard after login

Specific Features by User Type

- **Features of Admin**
 1. Manage Departments.
 2. Update fees/resources.
 3. Monitor queues and doctor performance.
- **Features of Doctor**
 1. View today queue.
 2. Refer patient to another department.
 3. Follow-up scheduling.
- **Features of Receptionist**
 1. Issue token for patient.
 2. Cancel/reschedule appointments.
 3. Prioritize on transfer patient.
- **Features of Patient**
 1. Check doctor availability.
 2. Track live queue position.
 3. View appointments and history.

N.B: Each user type must have at least 3 distinct features different from the others. Additional features may be added if desired. Proposals with generic, incomplete, or duplicated features across user types will receive reduced marks under the criteria below.